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IT362: Principles of Data Science

How AI is Reshaping Developer Language Choices Over Time

Logbook

Prepared By

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Phase	Task	Date	Description	Challenges
Phase 1: Data Collection, Research, and Assessment	Accessing GitHub API to collect data	September 18th	We started using python in google colab to apply the collecting code. We decided to collect 1000 observations for each year (250 per quarter) except for 2025 we ended up collecting almost 750 observations. The features we decided to collect for each repository were ["name", "owner", "url", "description", "stars", "forks", "created_at", "pushed_at", "primary_language", "all_languages_bytes","topics", "contributors_count"] to expand on the study.	API Rate Limits & Time-consuming collection, we solved this issues by dividing the task on the group members to get it done faster.
	Finalizing the raw data file	September 23rd	We gathered all the data collected by the group in one file and made sure they are compatible.	Merging all years' files together smoothly, we solved this issue by merging them via python then checking manually.
	Data checking and visualizing.	September 24th	We used python libraries such as numpy and matplotlib and seaborn to: -Check for any duplicate rows or missing values, there were no duplicates, but we found some missing values and we decided to only remove rows with missing "Primary programming language" value.	Hesitation regarding wither to remove all rows with any missing values or only the ones that misses "Primary programming language", eventually we decided to only remove rows with missing "Primary programming

				language" value only.
	Writing the 1st phase's report	September 22nd	We started writing the 1st draft for this phase's report.	No challenges faced.
Phase 2: Data Processing, Cleaning, and Exploratory Data Analysis (EDA)	Checking for dataset size	October 12th	We used statistically sufficient for analysis by calculating the required sample size to ensure the dataset has enough observations for the analysis. Applied the sample size formula $n = \frac{Z^2 * p * (1-p)}{e^2}$ using a confidence level of 95% and a 5% margin of error.	No challenges faced. The dataset was found to be sufficient for analysis, so we proceeded with it.
	Numerical Feature Transformation	October 13th	Applied Logarithmic Transformation (log(1+x)) to the stars and forks columns. This was crucial for handling the highly skewed distributions and mitigating the impact of extreme outliers, preparing the data for robust statistical modeling.	The main challenge was selecting the appropriate transformation technique to normalize the non-normal distributions of popularity metrics, which was addressed effectively by using the log(1+x) function
	Text and Temporal Feature Engineering	October 13th	Handled remaining missing values in text columns (description and topics) by replacing them with empty strings (' ') to allow for subsequent Text Vectorization. - Calculated the repository age in days (repo_age_days) from the 'created_at' timestamp for temporal analysis.	Ensuring that the missing text values were handled in a way that preserves the data structure
	Temporal Trend Analysis (Repository Creation)	October 13th	Analyzing the "Repository Creation Trend Over Time" graph. The analysis showed an increasing trend in the number of AI repository	Ensuring the date column was correctly formatted and grouped by month

			creations since 2020, with a noticeable rise in activity towards the end of 2024 and early 2025. This confirms the growing interest and activity in the AI/ML domain over the past few years.	(.dt.to_period("M")) to accurately plot the monthly trend was a critical step in this analysis.
	Temporal Trend Analysis (Average Popularity)	October 14th	Analyzed the "Average Stars per Month (2020-2025)" graph. The visualization shows a significant, sporadic increase in the average number of stars obtained by new repositories, particularly noting a sharp spike in late 2024/early 2025. This suggests that while more projects are being created, the average popularity of new projects has also seen periods of major growth.	The key challenge was interpreting the meaning of the spikes (which represent one or two highly starred projects in that month) versus the general baseline trend.
	Advanced EDA: Language Analysis Refinement	October 14th	- Refined the primary language analysis by merging 'Jupyter Notebook' entries with ‘Python’ to accurately represent the core programming language in AI projects. - Identified the most frequent primary languages and common co-occurring secondary languages (e.g., Shell, JavaScript) to understand project technology stacks.	The classification needed refinement to treat 'Jupyter Notebook' as an environment for 'Python' code rather than a standalone language.
	Advanced EDA: Topic & Correlation Analysis	October 14th	Analyzed the Top 10 most common topics associated with the AI/ML repositories (e.g., machine-learning, deep-learning, llm, openai). - Computed the correlation coefficient between ‘repo_age_days’ and ‘stars_log’, concluding that age has a very weak	The primary challenge was extracting meaningful insights from the repository topics, which was solved by focusing on high-frequency keyword analysis.

			correlation (0.028) with popularity.	
	Writing the 2nd phase's report	October 15th	We started writing the 2nd draft for this phase's report.	No challenges faced.
Phase 3: Advanced Modeling, Clustering, and Hypothesis Testing	Regression Data Preparation & Feature Engineering	November 17th	Prepared the Python-centric dataset by engineering new features: Extracted the **Top 10 Secondary Languages** (secondary_lang) used alongside Python and their corresponding byte counts (secondary_bytes), Applied Logarithmic Transformation (log(1+x)) to the secondary_bytes column to normalize the data and manage extreme outliers The most frequent secondary languages were confirmed as Shell (350 repos) and Jupyter Notebook (241 repos).	The all_languages_bytes column required careful parsing (using the ast library) to accurately extract the necessary secondary language information.
	Clustering Analysis (K-Means & DBSCAN)	November 17th	Performed clustering on normalized features (stars, forks, created_year, language_encoded) **K-Means:** Used K=4 based on the Elbow Method. The resulting Silhouette Score of **0.383** indicated moderately distinct clusters. **DBSCAN:** Failed to find meaningful clusters, identifying only **1** valid cluster with a noise ratio of 1.10%.	The moderate Silhouette Score from K-Means suggested that the chosen feature set was not sufficient to create sharply defined, natural groupings.
	Regression Modeling (Secondary Language Prediction)	November 19th	Trained four regression models (Baseline, Linear Regression, Decision Tree, and Random Forest) to predict the log-transformed secondary code size (log1p(secondary_bytes)).	The strong performance of tree-based models confirmed the non-linear and complex relationship between repository

			The **Random Forest Regressor** achieved the highest R^2 score of **0.912** and the lowest MSE of **2.11**, outperforming the Linear Regression model ($R^2=0.337$).	popularity metrics (stars/forks) and the volume of secondary codebase.
	Hypothesis Testing (ChatGPT Impact)	November 19th	Conducted two sample t-tests to evaluate the null hypothesis (H0): that the ChatGPT release (2022-11-30) had no significant impact on Python's role. Proportion of Python Usage: Observed a 7.45 percentage point decline (from 47.86% to 40.41%). Python Popularity (Stars): The average stars per Python repo surged by 168% (from 2,171 to 5,812). The T-Test yielded a P-value of 0.0000 leading to the **REJECTION of H0**	Ensuring the statistical significance of the star increase was accurately measured and interpreted as a quality-over-quantity shift.
	Temporal Analysis and Power Calculation	November 20th	Monthly Creation Rate: The T-Test on monthly repository counts FAILED TO REJECT H0 ($p=0.0633$). Statistical Power: Retrospective power analysis confirmed that the actual sample sizes were 4-7 times larger than required for high power. Statistical Power was 70.54%. suggesting the $p=0.063$ result may be a Type II error (missing a small, real effect).	Confirming the reliability of the statistical conclusions despite the borderline temporal result by validating that the samples were significantly large enough to detect the observed effects.
	Writing the Final Report and Conclusion	November 20th	We started writing the final report, synthesizing findings from all phases into a comprehensive conclusion that addresses the initial research objective.	No challenges faced